

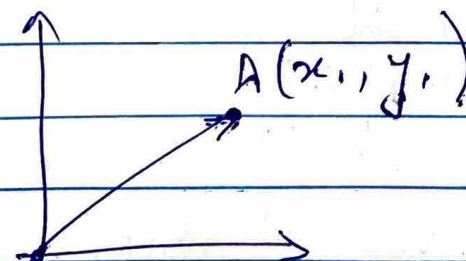
IR Basics

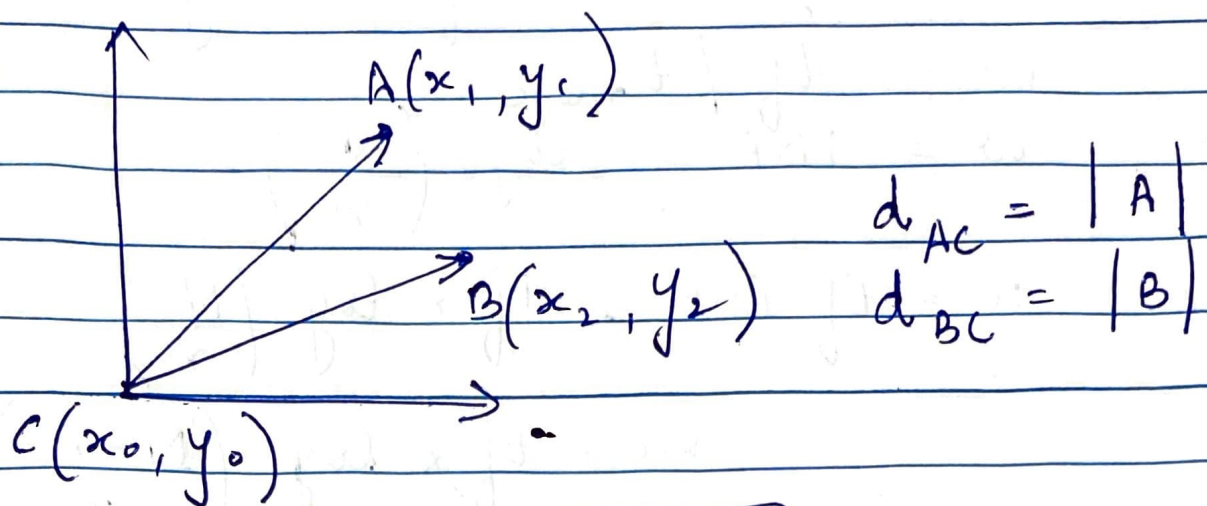
- markup free document text
- tokenization
- stopword removal
- stemming
- term weighing (based on counts)

	T_1	T_2	...	T_t	(terms)
D_1	d_{11}	d_{12}		d_{1t}	
D_2	d_{21}	d_{22}		d_{2t}	
D_n (docs)	d_{n1}	d_{n2}		d_{nt}	

- $A(x_1, y_1)$ & $B(x_2, y_2)$

dot product $\leftarrow A \cdot B = x_1 x_2 + y_1 y_2$

- 
distance $\leftarrow d = \sqrt{(x_1 - x_0)^2 + (y_1 - y_0)^2}$



$$\text{Sim}(A, B) = \cos \theta = \frac{A \cdot B}{|A| \cdot |B|}$$

$$= \frac{x_1 x_2 + y_1 y_2}{(x_1^2 + y_1^2)^{1/2} \cdot (x_2^2 + y_2^2)^{1/2}}$$

- This is one convenient way of ranking the docs by measuring how close their vectors are to a query vector. We compute the cosine angle btw. the query & each of the documents. Then, sort them in decreasing order of cosine angles (cosine similarity).
- by defining, $t_{\max}$ as max. term freq. in a document, N as no. of docs in a collecⁿ & n as no. of docs containing query term, we redefine the term wts as —

→ term frequency

$$w = tf / t_{\max}$$

$$w = IDF = \log \left(\frac{N}{n} \right)$$

$$w = tf \cdot IDF = tf \times \log \left(\frac{N}{n} \right)$$

$$w = tf \times \log \left(\frac{N-n}{n} \right)$$

- text similarity

$$\text{sim}(Q, D_i) = \sum w_{q,j} w_{i,j} \quad \text{→ dot prod.}$$

$$\sqrt{\sum_j w_{q,j}^2} \quad \sqrt{\sum_i w_{i,j}^2}$$

← mag.
of query

← magnitude
of ith doc

→ to do, we construct a term space: It is defined by a list of terms. These terms are extracted from the collecⁿ of docs to be queried. The coordinates of these pts. are acc. to the weighting scheme.

$$w_i = tf_i \times \log \left(\frac{D}{df_i} \right)$$

t_{fi} $\Rightarrow$ term freq. or no. of times a term i occurs in a doc.

d_{fi} $\Rightarrow$ no. of docs containing term i

D $\Rightarrow$ no. of docs in DB.

- few things to observe -

$\rightarrow$ very frequent terms tend to receive low wts (~ 0)

$\rightarrow$ model correctly predicts the very common terms occurring in docs

$\rightarrow$ they are not good discriminators of relevancy.

$\rightarrow$ instead of indi. vectors lens & dot products, use same computational time by directly applying the similarity funcⁿ.

- limitations -

$\rightarrow$ calculation intensive; high process time

$\rightarrow$ each time a new term is added, we recalculate all vectors.

$\rightarrow$ order of the terms is lost vector space representaⁿ.

$\rightarrow$ very long docs make similarity measure difficult.

$\rightarrow$ false negative matches
or positive

→ sys. for handling semantic content may use special tags.

- how to improve the model?

→ get a set of keywords representing each doc.

→ eliminate all stop words & very frequently occurring terms

→ stemming terms to roots

→ limit the vector space to nouns, verbs, & descriptive adjectives

→ use small signature files

→ compute subvectors in large docs

→ don't retrieve docs below certain threshold

- Polysemy → Terms can be used to express diff. things in diff. context.

Some docs (irrelevant) may have high similarity because of this. It affects precision.

- Synonymity → Terms used to express the same thing. This affects recall.

- Currently, no search engine use tf. idf. As it is easy to decrease by tf term.

- Keyword spam - A spammer only need to repeat the keyword many times to inc. the wt.
- Also, term vector models assume term independence, which is not true
- Normalized Wts.

$$w_{i,j} = \frac{t_{f_{i,j}}}{\max_j t_{f_{i,j}}} \log \left(\frac{D}{df_i} \right)$$

- Glasgow model

$$w_{ij} = \frac{\log(f_{ij} + 1)}{\log(len_j)} \cdot \log \left(\frac{N}{n_i} \right)$$

f_{ij} → freq. of term i in doc j
 len_j → no. of unique terms in doc j
 N → total docs.
 n_i → total docs. containing term i .
 $t_{f_{i,j}}$ → idf wt.

- Precision = $\frac{TP}{TP + FP}$ of all ^{models} correct, how much actually correct

- Recall = $\frac{TP}{TP + FN}$ of all actually correct, how many model said correct.